



Armada Kalamunda Group

Non-Invasive Ventilation (Acute Management) Guideline

1. Purpose

This document provides information about non-invasive ventilation (NIV) to enable healthcare professionals to make appropriate decisions for individual patients.

2. Applicability

This guideline applies to medical, nursing and allied health staff caring for patients admitted to the Intensive Care Unit (ICU) requiring NIV.

This guideline excludes devices used for sleep apnoea and outpatient services and does not consider the use of NIV in the post-operative setting or as an adjunct to weaning from invasive ventilation.

3. Background

NIV, the provision of assisted ventilation without the use of an artificial airway, is now an accepted and effective therapy for patients with acute respiratory failure (ARF).

There is good evidence that it is the preferred first-line therapy in selected patients with chronic obstructive pulmonary disease (COPD) and acute pulmonary oedema.

It has the advantages of avoidance of an artificial airway which improves patient comfort, reduction in the need for sedation and preservation of the natural defences of the upper airway.

Patients must be nursed on 1:1 ratio on initiation of NIV

Patients may be nursed on 1:2 ratio once stabilised and only after reassessment of their overall clinical condition by the clinical coordinator and medical staff

4. Guidelines

4.1 Patient selection

The best evidence for benefit is in patients with

- ARF related to an exacerbation of COPD
- Cardiogenic pulmonary oedema
- Acute Hypoxaemia respiratory failure
- Post extubation respiratory failure
- Pneumonia in an immunocompromised patient

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Predictors of NIV failure include greater severity of acute illness, previous functional limitation from COPD, comorbidities, older age, failure to respond quickly and technical problems (mask leak, breathing coordination). Patients with community-acquired pneumonia often have a poor response to NIV.

NIV should be **avoided** in patients with

- An inability to protect the airway (impaired consciousness) or to clear secretions
- Haemodynamic instability
- Airway obstruction
- Gastrointestinal bleeding or vomiting, or recent gastro-oesophageal surgery
- Facial surgery or other reason why a facial device cannot be effective.

4.2 Contraindications for procedure

Precautionary criteria	Relative
• Untreated pneumothorax • Imminent Cardiac/Respiratory Arrest • Unstable multiple organ failure • Haemodynamic instability • CSF leak / Base of skull fracture • Anatomical obstruction of upper airway • Gastrointestinal bleeding or ileus • Lack of spontaneous breathing	• Poor bulbar function / high risk of aspiration • Vomiting • Unco-operative or unmotivated patients • Extensive facial lacerations/unstable fractures or burns • Inability to effectively clear secretions • Recent oesophageal / lung surgery • Inability to fit an acceptable mask

4.3 General care

NIV should only be initiated by a medical officer experienced in its use and is only performed in the Emergency Department or the Intensive Care Unit.

Patient supervision and observation should include:

- Continuous pulse oximetry and cardiac monitoring (hourly)
- Respiratory effort, chest expansion and respiratory rate (hourly)
- Airway patency
- Blood pressure and heart rate (hourly)
- Level of consciousness.
- There should be close attendance of medical staff from initiation of NIV until patient is stable to ensure that patient compliance, mask fit, comfort and synchrony with the ventilator is managed appropriately
- Tidal volume, minute ventilation, and peak airway pressure (hourly).
- Chest auscultation should be documented at the beginning of each shift and a minimum of 2 hourly.

- Ventilator settings should be recorded each hour or when a change to settings occurs. These should include Mode, FiO₂, and inspiratory and or expiratory pressures set.
- Monitoring of the patients Arterial Blood Gases (ABG) and/or Venous Blood Gases (VBG) is used to guide treatment and predict outcomes.
- As a minimum ABG's/VBGs should be obtained and reviewed by the MO at 1/24 , 4/24 and as clinically indicated on all patients requiring NIV

4.4 Ongoing management following NIV initiation

- The efficacy of NIV should be reviewed as required, but at least daily by medical staff to decide whether to continue or escalate to invasive ventilation.
- The NIV mask should be removed at intervals, as tolerated by the patient's clinical condition, to facilitate activities such as meals or mouth care. Consider supplementary O₂ requirements during these periods.
- Oxygen (O₂) therapy is to be administered as prescribed by MO.
- Consider intravenous hydration/enteral feeding if the patient is unable to tolerate removal of NIV.
- Monitor fluid intake and output on fluid balance chart.
- Assess for signs of abdominal distension, auscultate abdomen for bowel sounds, if gastric distension is present discuss with MO.
- Check for pressure areas from the mask and/or head straps.
- Patients should be nursed 1:1 on initiation of NIV and may be nursed 1:2 once stabilised after reassessment of clinical requirements by the nursing unit coordinator and medical staff.
- Patients with quadriplegia or physically restrained upper limbs, will require a companion or nursing special to be present to be able to remove the mask should they require to. E.g. If they need to vomit.
- Patients on NIV should be visible to a nursing staff or allied healthcare worker at all times.

4.5 Potential problems during NIV or post procedure

- | | |
|-------------------------------------|-------------------------|
| • Air leaks | • Aspiration |
| • Mask discomfort | • Gastric distension |
| • Eye irritation | • Hypotension |
| • Sinus congestion | • Pulmonary barotrauma |
| • Oro-nasal drying | • Facial pressure areas |
| • Patient / ventilator dyssynchrony | |

4.6 Initiation of NIV

- Obtain and document a full set of physiological observations including level of consciousness and respiratory assessment prior to commencing NIV therapy.
- Explain the procedure/s to the patient, family and/or carer and gain appropriate consent as able.
- Perform the ventilator and patient circuit check as per manufacturer's instructions.
- Provide rationale for keeping the mask on to improve compliance with therapy
- If possible, a full face mask should be used on initiation.

- It is essential that the mask can be removed quickly in case of vomiting to avoid aspiration of gastric contents
- Add humidification if possible as per manufacturer's guidelines.
- Where possible the patient should be in a sitting or semi-recumbent position to enhance chest expansion. Alternate positions may be recommended based on clinical condition.
- Patients requiring NIV outside of an Emergency Department or Intensive Care Unit should be referred to Intensive Care team for review.

4.7 Weaning of NIV in the ED

- When patients present acutely to the ED, NIV is often utilised to alleviate respiratory distress. If the cause is cardiogenic pulmonary oedema these patients can sometimes be weaned from NIV in a short space of time, when other measures such as diuresis or heart rate control have had time to take effect.
- In patients in whom the cause of respiratory distress is COPD, definitive therapies (bronchodilators, steroids) take longer to achieve a benefit and weaning from NIV may not be possible in a timeframe which allows for good flow of ED patients. In this group of patients, an ICU opinion should be requested about the prospects of weaning in a short timeframe; and if this is not thought to be possible, ICU admission should be discussed.

5. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Consent to Treatment Policy MP 0175/22](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching EMHS: 21](#)

6. Supporting information

The following documents support the implementation of this policy:

- [Oxygen Therapy Management in Adults Guideline AKG-GUI-0335-17](#)
- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14](#)

7. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.5 Healthcare rights and informed consent
- Standard 6 – Action 6.6 Correct identification and procedure matching
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- Standard 8 – Action 8.3 Partnering with consumers

- Standard 8 – Action 8.4 Recognising acute deterioration
- Standard 8 – Action 8.10 Responding to deterioration

9. Document owner

Head of Intensive Care Unit

Enquiries relating to this guideline may be directed to:

Title: Head of Department

Department: Intensive Care Unit

Email:

10. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V6	03/10/2024	03/10/2027	Scheduled review

11. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
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